CS480P/580P Introduction to Natural Language Processing

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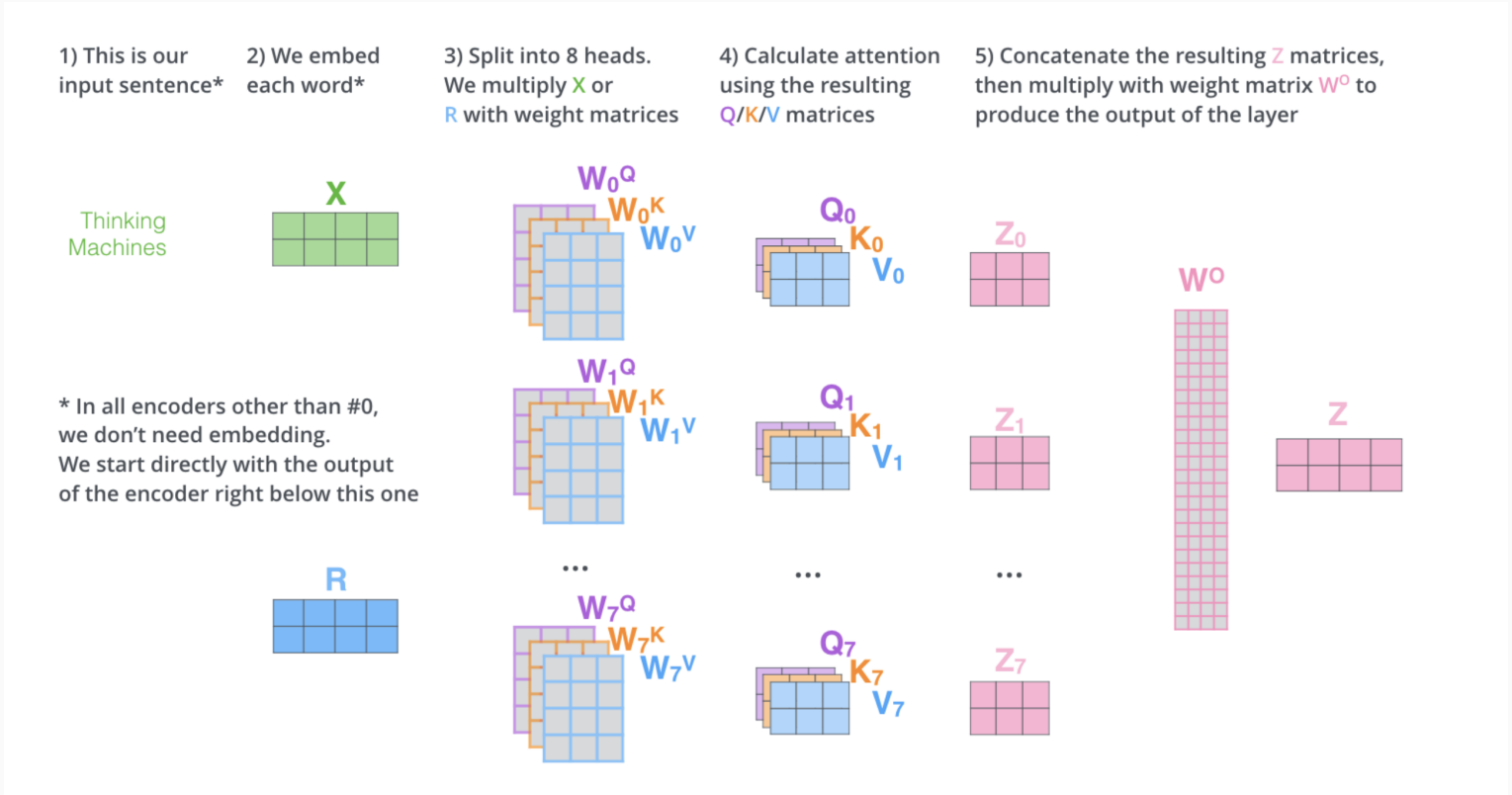
Lecture 12

Overall architecture



TL;DR: You have to play around with model.py yourself

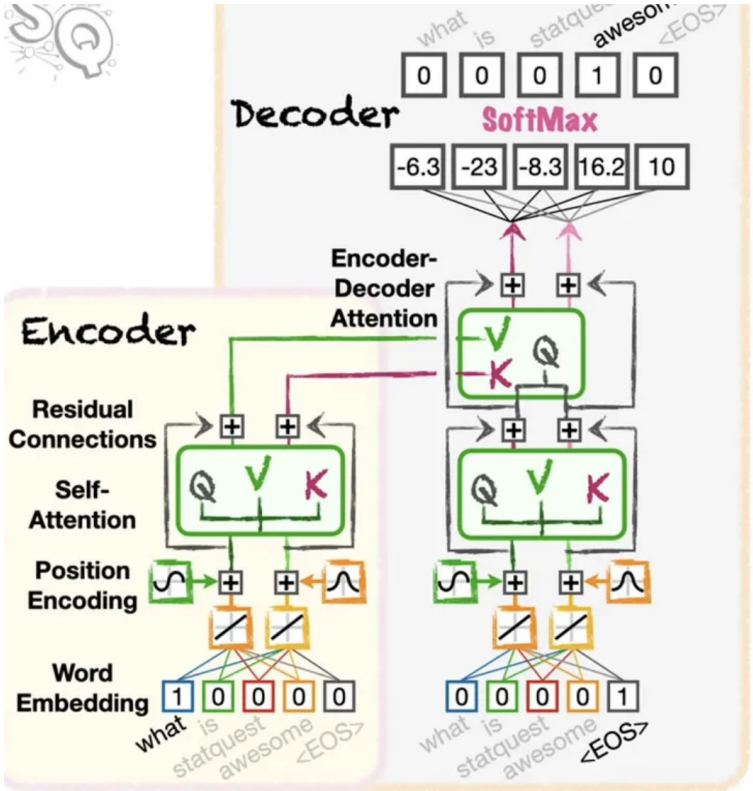
minGPT / mingpt / **model.py**

mishig25 Use XOR operator ^ for checking assertion `type_given XOR params_gi...

Code | Blame | 310 lines (272 loc) · 14.3 KB

```
"""
Full definition of a GPT Language Model, all of it in this single file.

References:
1) the official GPT-2 TensorFlow implementation released by OpenAI:
https://github.com/openai/gpt-2/blob/master/src/model.py
2) huggingface/transformers PyTorch implementation:
https://github.com/huggingface/transformers/blob/main/src/transformers/models/gpt2/modeling_gpt2.py
"""

import math

import torch
import torch.nn as nn
from torch.nn import functional as F

from mingpt.utils import CfgNode as CN

# -----------------------------------------------------------------------------

class NewGELU(nn.Module):
    """
    Implementation of the GELU activation function currently in Google BERT repo (identical to OpenAI GPT).
    Reference: Gaussian Error Linear Units (GELU) paper: https://arxiv.org/abs/1606.08415
    """
    def forward(self, x):
        return 0.5 * x * (1.0 + torch.tanh(math.sqrt(2.0 / math.pi) * (x + 0.044715 * torch.pow(x, 3.0))))

class CausalSelfAttention(nn.Module):
    """
```

Encoder-Decoder, Encoder-Only, Decoder-Only



A normal **Transformer** uses one unit to encode the input, called the **Encoder**, and a separate unit to generate the output, called the **Decoder**.

And a normal **Transformer** uses two types of **Attention** during inference: **Self-Attention** and **Encoder-Decoder Attention**.

Lastly, during **Training**, a normal **Transformer** uses **Masked Self-Attention**, but only on the output.

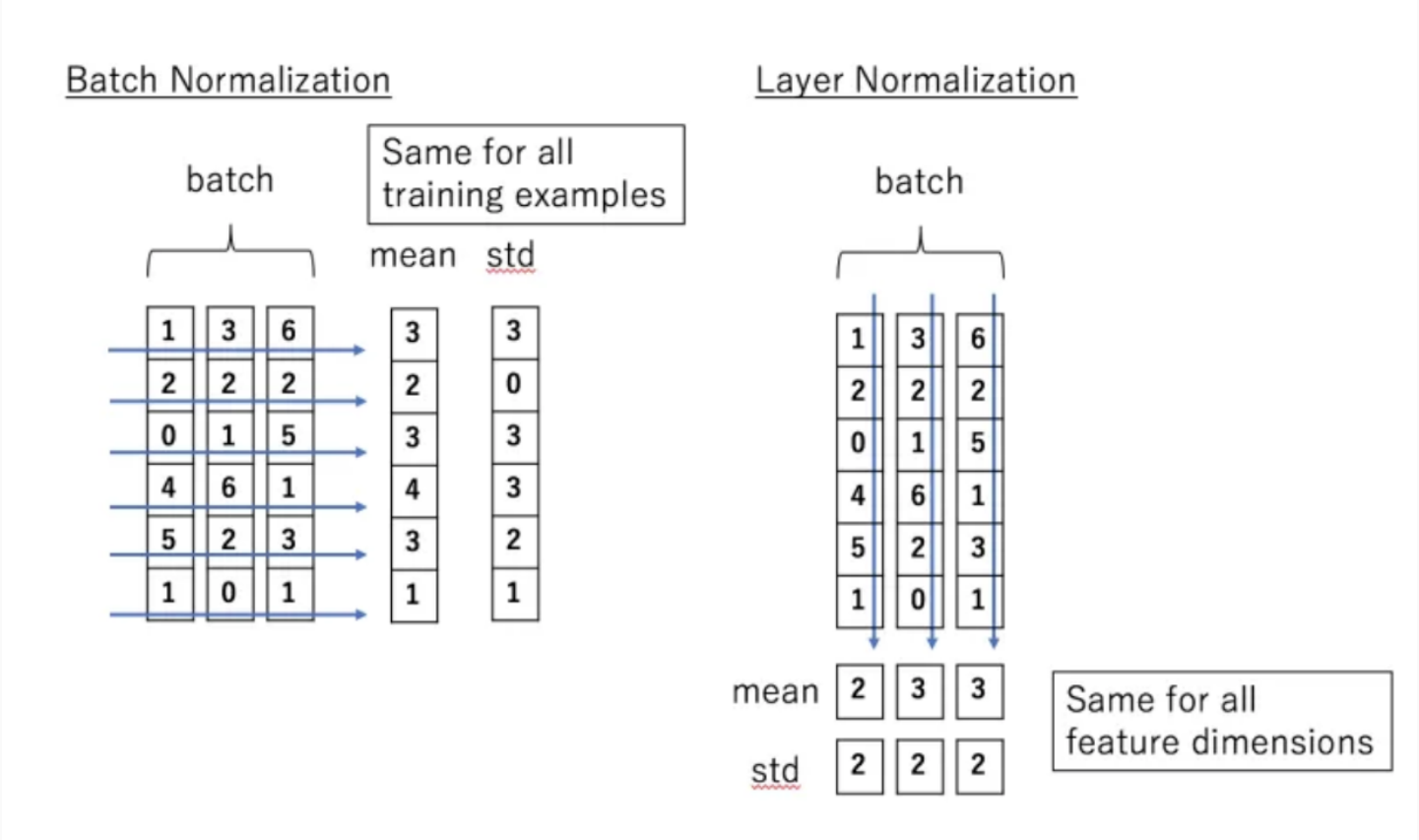
transformer

```
self.transformer = nn.ModuleDict(dict(
    wte = nn.Embedding(config.vocab_size, config.n_embd),
    wpe = nn.Embedding(config.block_size, config.n_embd),
    drop = nn.Dropout(config.embd_pdrop),
    h = nn.ModuleList([Block(config) for _ in range(config.n_layer)]),
    ln_f = nn.LayerNorm(config.n_embd),
))
```

LayerNorm BatchNorm

1. As the depth of the network increases, the distribution of feature values in each layer will gradually approach the upper and lower ends of the activation function's output range, causing the activation function to saturate. Continuing in this way can lead to gradient vanishing. Normalization can recenter the distribution of feature values to a standard normal distribution, ensuring that the feature values fall within the range where the activation function is more sensitive to inputs.
2. We assume Independent and Identically Distributed (IID), which assumes that the training data and test data follow the same distribution. Normalizing the training and test data can prevent the influence of different data distributions on the model.

LayerNorm BatchNorm



https://medium.com/@florian_*algo*/batchnorm-and-layernorm-2637f46a998b

LayerNorm BatchNorm

CLASS torch.nn.LayerNorm(*normalized_shape*, *eps=1e-05*, *elementwise_affine=True*, *bias=True*, *device=None*, *dtype=None*) [SOURCE]

Applies Layer Normalization over a mini-batch of inputs.

This layer implements the operation as described in the paper Layer Normalization

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}} * \gamma + \beta$$

The mean and standard-deviation are calculated over the last *D* dimensions, where *D* is the dimension of `normalized_shape`. For example, if `normalized_shape` is `(3, 5)` (a 2-dimensional shape), the mean and standard-deviation are computed over the last 2 dimensions of the input (i.e. `input.mean((-2, -1))`). γ and β are learnable affine transform parameters of `normalized_shape` if `elementwise_affine` is `True`. The standard-deviation is calculated via the biased estimator, equivalent to *torch.var(input, unbiased=False)*.

transformer

```
self.transformer = nn.ModuleDict(dict(
    wte = nn.Embedding(config.vocab_size, config.n_embd),
    wpe = nn.Embedding(config.block_size, config.n_embd),
    drop = nn.Dropout(config.embd_pdrop),
    h = nn.ModuleList([Block(config) for _ in range(config.n_layer)]),
    ln_f = nn.LayerNorm(config.n_embd),
))
```

generate

```
[ ]  device = 'cpu'
     prompt = "Binghamton University, the"
     tokenizer = BPETokenizer()
     x = tokenizer(prompt).to(device)
```

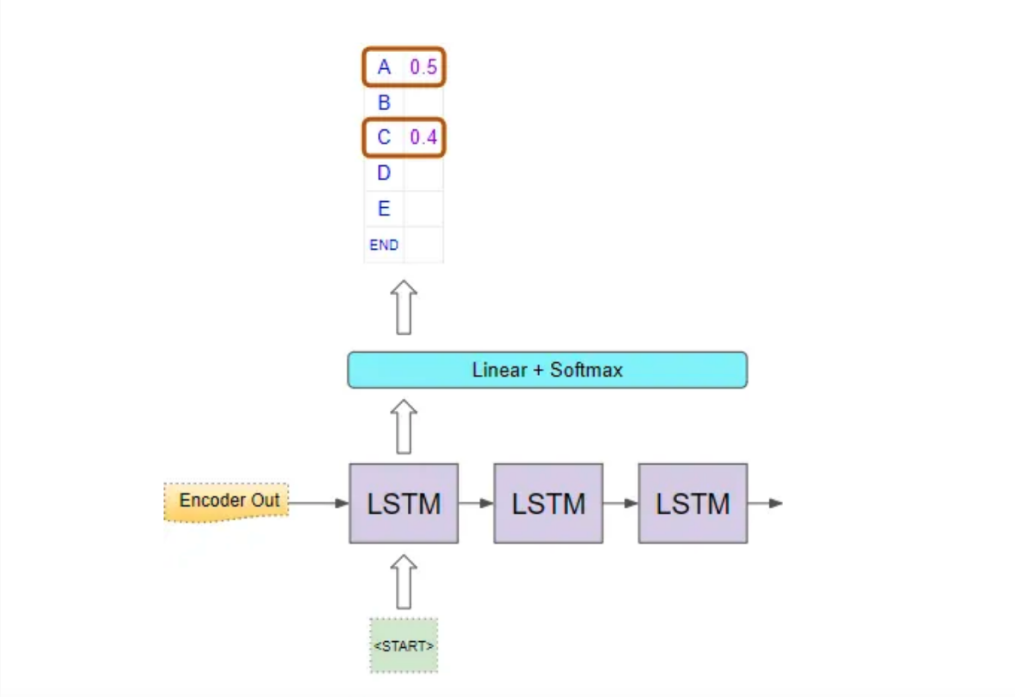
```
     downloading https://openaipublic.blob.core.windows.net/gpt-2/models/124M/encoder.json to /root/.cache/mingpt/encoder.json
     downloading https://openaipublic.blob.core.windows.net/gpt-2/models/124M/vocab.bpe to /root/.cache/mingpt/vocab.bpe
```

```
[ ]  y = model.generate(x, max_new_tokens=40, do_sample=False, top_k=40)
```

generate: which line is the main inference of the model?

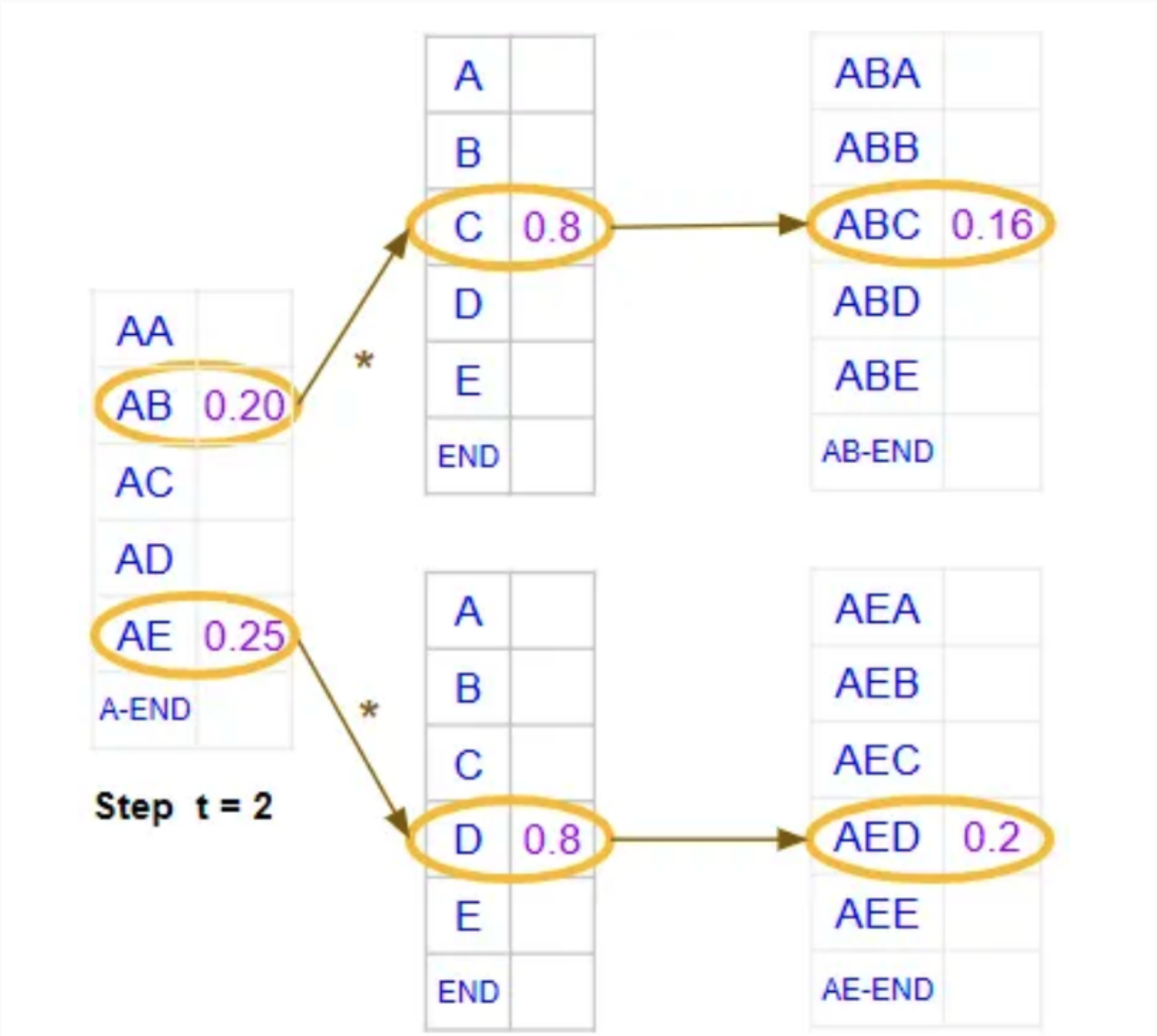
```python
def generate(self, idx, max_new_tokens, temperature=1.0, do_sample=False, top_k=None):
    """
    Take a conditioning sequence of indices idx (LongTensor of shape (b,t)) and complete
    the sequence max_new_tokens times, feeding the predictions back into the model each time.
    Most likely you'll want to make sure to be in model.eval() mode of operation for this.
    """
    for _ in range(max_new_tokens):
        # if the sequence context is growing too long we must crop it at block_size
        idx_cond = idx if idx.size(1) <= self.block_size else idx[:, -self.block_size:]
        # forward the model to get the logits for the index in the sequence
        logits, _ = self(idx_cond)
        # pluck the logits at the final step and scale by desired temperature
        logits = logits[:, -1, :] / temperature
        # optionally crop the logits to only the top k options
        if top_k is not None:
            v, _ = torch.topk(logits, top_k)
            logits[logits < v[:, [-1]]] = -float('Inf')
        # apply softmax to convert logits to (normalized) probabilities
        probs = F.softmax(logits, dim=-1)
        # either sample from the distribution or take the most likely element
        if do_sample:
            idx_next = torch.multinomial(probs, num_samples=1)
        else:
            _, idx_next = torch.topk(probs, k=1, dim=-1)
        # append sampled index to the running sequence and continue
        idx = torch.cat((idx, idx_next), dim=1)

    return idx
```

Beam Search

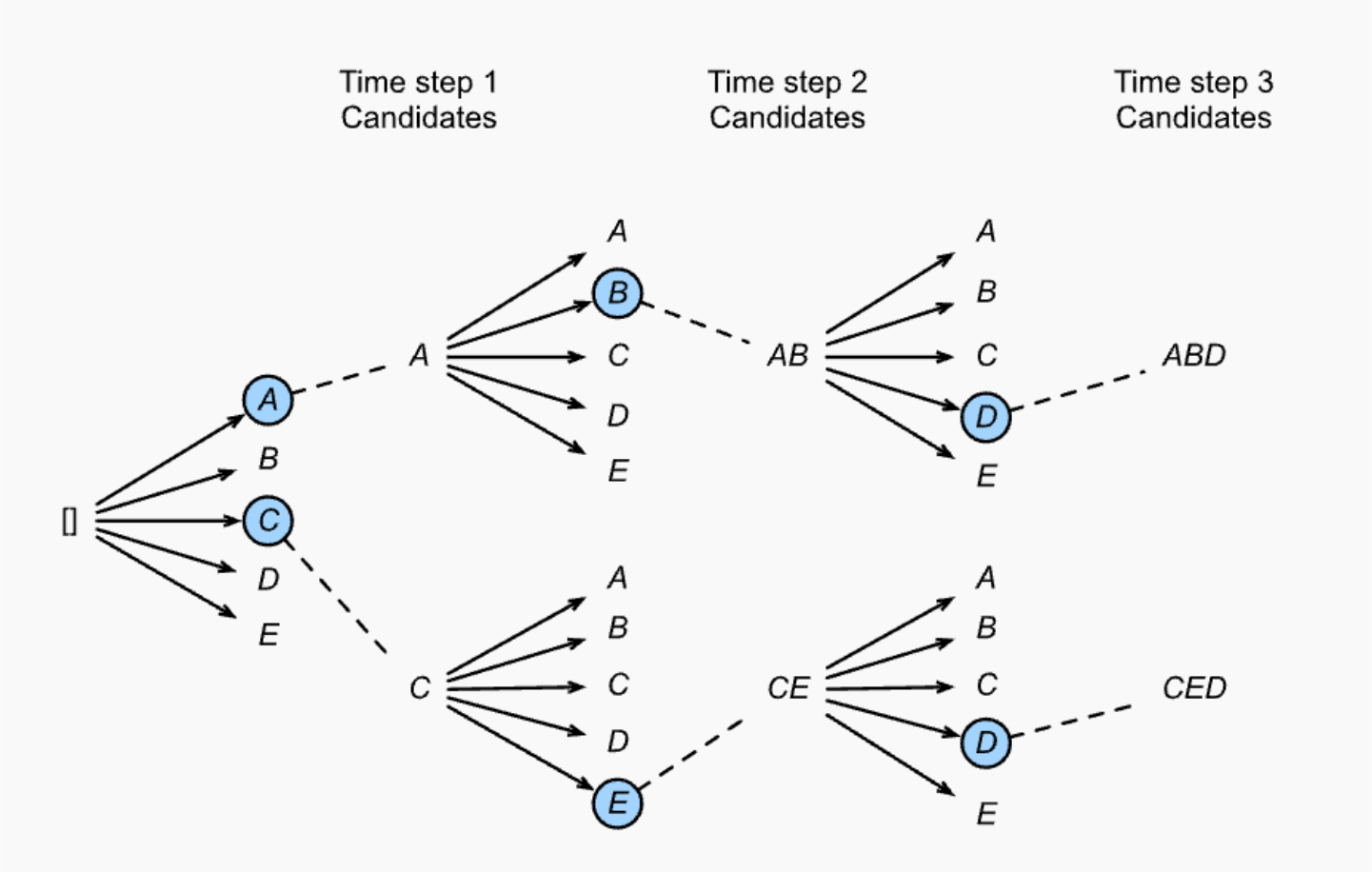


- https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24
- https://d2l.ai/chapter_recurrent-modern/beam-search.html

Beam Search


AA
AB 0.20
AC
AD
AE 0.25
A-END
Step t = 2
*
*
A
B
C 0.8
D
E
END
A
B
C
D 0.8
E
END
ABA
ABB
ABC 0.16
ABD
ABE
AB-END
AEA
AEB
AEC
AED 0.2
AEE
AE-END

Beam Search


Time step 1
Candidates
Time step 2
Candidates
Time step 3
Candidates
[]
A
B
C
D
E
A
B
C
D
E
AB
CE
ABD
CED

generate: What's max new tokens?

```
def generate(self, idx, max_new_tokens, temperature=1.0, do_sample=False, top_k=None):
    """
    Take a conditioning sequence of indices idx (LongTensor of shape (b,t)) and complete
    the sequence max_new_tokens times, feeding the predictions back into the model each time.
    Most likely you'll want to make sure to be in model.eval() mode of operation for this.
    """
    for _ in range(max_new_tokens):
        # if the sequence context is growing too long we must crop it at block_size
        idx_cond = idx if idx.size(1) <= self.block_size else idx[:, -self.block_size:]
        # forward the model to get the logits for the index in the sequence
        logits, _ = self(idx_cond)
        # pluck the logits at the final step and scale by desired temperature
        logits = logits[:, -1, :] / temperature
        # optionally crop the logits to only the top k options
        if top_k is not None:
            v, _ = torch.topk(logits, top_k)
            logits[logits < v[:, [-1]]] = -float('Inf')
        # apply softmax to convert logits to (normalized) probabilities
        probs = F.softmax(logits, dim=-1)
        # either sample from the distribution or take the most likely element
        if do_sample:
            idx_next = torch.multinomial(probs, num_samples=1)
        else:
            _, idx_next = torch.topk(probs, k=1, dim=-1)
        # append sampled index to the running sequence and continue
        idx = torch.cat((idx, idx_next), dim=1)

    return idx
```

generate: What's the decoding scheme of this method?

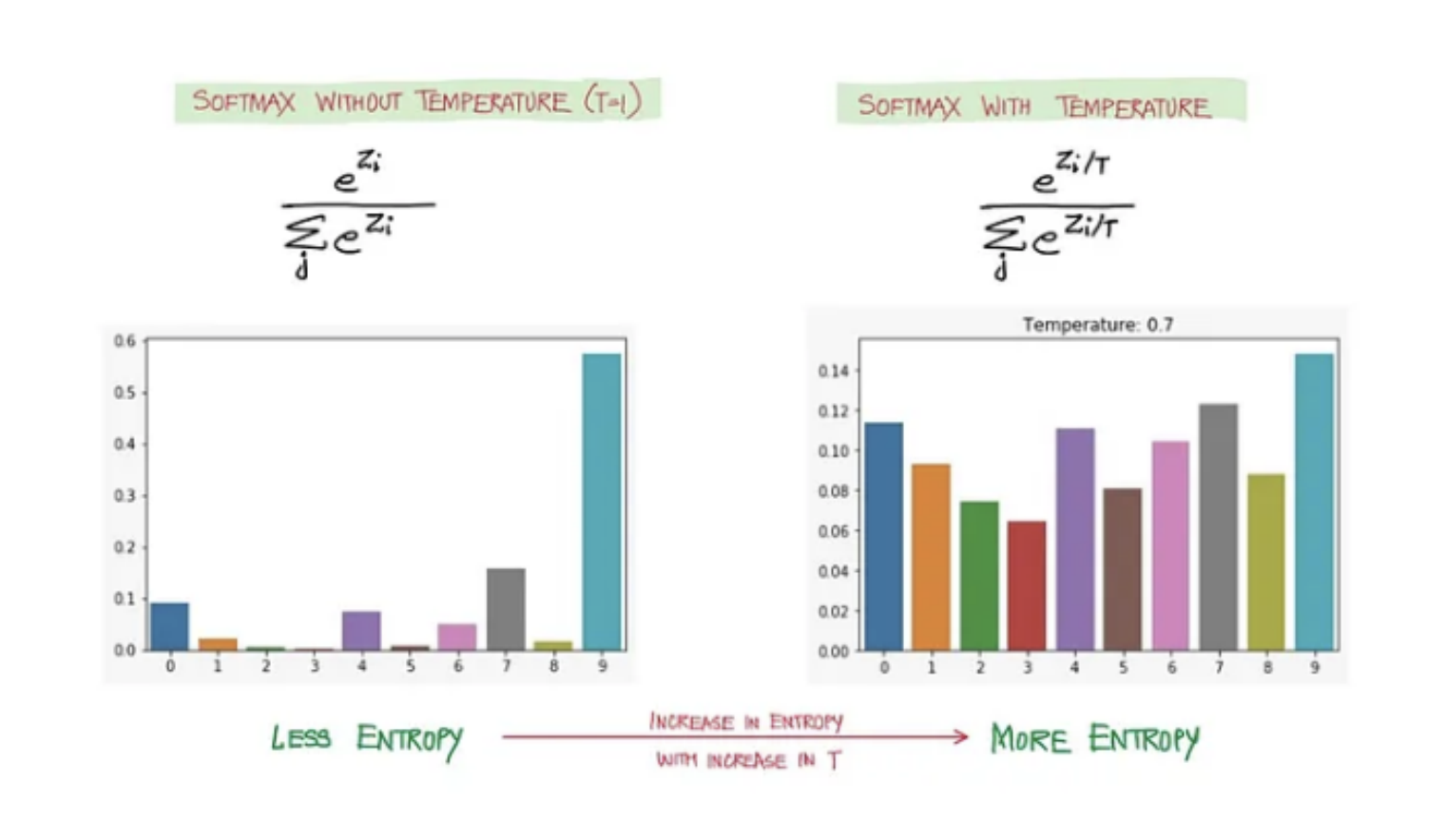
```
def generate(self, idx, max_new_tokens, temperature=1.0, do_sample=False, top_k=None):
    """
    Take a conditioning sequence of indices idx (LongTensor of shape (b,t)) and complete
    the sequence max_new_tokens times, feeding the predictions back into the model each time.
    Most likely you'll want to make sure to be in model.eval() mode of operation for this.
    """
    for _ in range(max_new_tokens):
        # if the sequence context is growing too long we must crop it at block_size
        idx_cond = idx if idx.size(1) <= self.block_size else idx[:, -self.block_size:]
        # forward the model to get the logits for the index in the sequence
        logits, _ = self(idx_cond)
        # pluck the logits at the final step and scale by desired temperature
        logits = logits[:, -1, :] / temperature
        # optionally crop the logits to only the top k options
        if top_k is not None:
            v, _ = torch.topk(logits, top_k)
            logits[logits < v[:, [-1]]] = -float('Inf')
        # apply softmax to convert logits to (normalized) probabilities
        probs = F.softmax(logits, dim=-1)
        # either sample from the distribution or take the most likely element
        if do_sample:
            idx_next = torch.multinomial(probs, num_samples=1)
        else:
            _, idx_next = torch.topk(probs, k=1, dim=-1)
        # append sampled index to the running sequence and continue
        idx = torch.cat((idx, idx_next), dim=1)

    return idx
```

generate: Do you remember what's temperature?

```python
def generate(self, idx, max_new_tokens, temperature=1.0, do_sample=False, top_k=None):
    """
    Take a conditioning sequence of indices idx (LongTensor of shape (b,t)) and complete
    the sequence max_new_tokens times, feeding the predictions back into the model each time.
    Most likely you'll want to make sure to be in model.eval() mode of operation for this.
    """
    for _ in range(max_new_tokens):
        # if the sequence context is growing too long we must crop it at block_size
        idx_cond = idx if idx.size(1) <= self.block_size else idx[:, -self.block_size:]
        # forward the model to get the logits for the index in the sequence
        logits, _ = self(idx_cond)
        # pluck the logits at the final step and scale by desired temperature
        logits = logits[:, -1, :] / temperature
        # optionally crop the logits to only the top k options
        if top_k is not None:
            v, _ = torch.topk(logits, top_k)
            logits[logits < v[:, [-1]]] = -float('Inf')
        # apply softmax to convert logits to (normalized) probabilities
        probs = F.softmax(logits, dim=-1)
        # either sample from the distribution or take the most likely element
        if do_sample:
            idx_next = torch.multinomial(probs, num_samples=1)
        else:
            _, idx_next = torch.topk(probs, k=1, dim=-1)
        # append sampled index to the running sequence and continue
        idx = torch.cat((idx, idx_next), dim=1)

    return idx
```

Softmax Temperature

- $\frac{exp(z_i/T)}{\sum exp(z_i/T)}$
- https://medium.com/mlearning-ai/softmax-temperature-5492e4007f71



forward: What is cross entroy trying to do?

```python
def forward(self, idx, targets=None):
    device = idx.device
    b, t = idx.size()
    assert t <= self.block_size, f"Cannot forward sequence of length {t}, block size is only {self.block_size}"
    pos = torch.arange(0, t, dtype=torch.long, device=device).unsqueeze(0) # shape (1, t)

    # forward the GPT model itself
    tok_emb = self.transformer.wte(idx) # token embeddings of shape (b, t, n_embd)
    pos_emb = self.transformer.wpe(pos) # position embeddings of shape (1, t, n_embd)
    x = self.transformer.drop(tok_emb + pos_emb)
    for block in self.transformer.h:
        x = block(x)
    x = self.transformer.ln_f(x)
    logits = self.lm_head(x)

    # if we are given some desired targets also calculate the loss
    loss = None
    if targets is not None:
        loss = F.cross_entropy(logits.view(-1, logits.size(-1)), targets.view(-1), ignore_index=-1)

    return logits, loss
```

$$L_{CE} = -\sum_{i=1}^{V} y_i log(p_i)$$

forward: If I tell you this is a LM, dimensionality of lmhead?

```python
def forward(self, idx, targets=None):
    device = idx.device
    b, t = idx.size()
    assert t <= self.block_size, f"Cannot forward sequence of length {t}, block size is only {self.block_size}"
    pos = torch.arange(0, t, dtype=torch.long, device=device).unsqueeze(0) # shape (1, t)

    # forward the GPT model itself
    tok_emb = self.transformer.wte(idx) # token embeddings of shape (b, t, n_embd)
    pos_emb = self.transformer.wpe(pos) # position embeddings of shape (1, t, n_embd)
    x = self.transformer.drop(tok_emb + pos_emb)
    for block in self.transformer.h:
        x = block(x)
    x = self.transformer.ln_f(x)
    logits = self.lm_head(x)

    # if we are given some desired targets also calculate the loss
    loss = None
    if targets is not None:
        loss = F.cross_entropy(logits.view(-1, logits.size(-1)), targets.view(-1), ignore_index=-1)

    return logits, loss
```

$$L_{CE} = -\sum_{i=1}^{V} y_i log(p_i)$$

forward: What's the shape of token embedding?

```python
def forward(self, idx, targets=None):
    device = idx.device
    b, t = idx.size()
    assert t <= self.block_size, f"Cannot forward sequence of length {t}, block size is only {self.block_size}"
    pos = torch.arange(0, t, dtype=torch.long, device=device).unsqueeze(0) # shape (1, t)

    # forward the GPT model itself
    tok_emb = self.transformer.wte(idx) # token embeddings of shape (b, t, n_embd)
    pos_emb = self.transformer.wpe(pos) # position embeddings of shape (1, t, n_embd)
    x = self.transformer.drop(tok_emb + pos_emb)
    for block in self.transformer.h:
        x = block(x)
    x = self.transformer.ln_f(x)
    logits = self.lm_head(x)

    # if we are given some desired targets also calculate the loss
    loss = None
    if targets is not None:
        loss = F.cross_entropy(logits.view(-1, logits.size(-1)), targets.view(-1), ignore_index=-1)

    return logits, loss
```

$$L_{CE} = -\sum_{i=1}^{V} y_i log(p_i)$$

forward: What's the shape of position embedding?

```python
def forward(self, idx, targets=None):
    device = idx.device
    b, t = idx.size()
    assert t <= self.block_size, f"Cannot forward sequence of length {t}, block size is only {self.block_size}"
    pos = torch.arange(0, t, dtype=torch.long, device=device).unsqueeze(0) # shape (1, t)

    # forward the GPT model itself
    tok_emb = self.transformer.wte(idx) # token embeddings of shape (b, t, n_embd)
    pos_emb = self.transformer.wpe(pos) # position embeddings of shape (1, t, n_embd)
    x = self.transformer.drop(tok_emb + pos_emb)
    for block in self.transformer.h:
        x = block(x)
    x = self.transformer.ln_f(x)
    logits = self.lm_head(x)

    # if we are given some desired targets also calculate the loss
    loss = None
    if targets is not None:
        loss = F.cross_entropy(logits.view(-1, logits.size(-1)), targets.view(-1), ignore_index=-1)

    return logits, loss
```

$$L_{CE} = -\sum_{i=1}^{V} y_i log(p_i)$$

transformer

```
self.transformer = nn.ModuleDict(dict(
    wte = nn.Embedding(config.vocab_size, config.n_embd),
    wpe = nn.Embedding(config.block_size, config.n_embd),
    drop = nn.Dropout(config.embd_pdrop),
    h = nn.ModuleList([Block(config) for _ in range(config.n_layer)]),
    ln_f = nn.LayerNorm(config.n_embd),
))
```

Block: What is block doing?

```python
class Block(nn.Module):
    """ an unassuming Transformer block """

    def __init__(self, config):
        super().__init__()
        self.ln_1 = nn.LayerNorm(config.n_embd)
        self.attn = CausalSelfAttention(config)
        self.ln_2 = nn.LayerNorm(config.n_embd)
        self.mlp = nn.ModuleDict(dict(
            c_fc    = nn.Linear(config.n_embd, 4 * config.n_embd),
            c_proj  = nn.Linear(4 * config.n_embd, config.n_embd),
            act     = NewGELU(),
            dropout = nn.Dropout(config.resid_pdrop),
        ))
        m = self.mlp
        self.mlpf = lambda x: m.dropout(m.c_proj(m.act(m.c_fc(x)))) # MLP forward

    def forward(self, x):
        x = x + self.attn(self.ln_1(x))
        x = x + self.mlpf(self.ln_2(x))
        return x
```

Causal Attention

```
def forward(self, x):
    B, T, C = x.size() # batch size, sequence length, embedding dimensionality (n_embd)

    # calculate query, key, values for all heads in batch and move head forward to be the batch dim
    q, k ,v  = self.c_attn(x).split(self.n_embd, dim=2)
    k = k.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)
    q = q.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)
    v = v.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)

    # causal self-attention; Self-attend: (B, nh, T, hs) x (B, nh, hs, T) -> (B, nh, T, T)
    att = (q @ k.transpose(-2, -1)) * (1.0 / math.sqrt(k.size(-1)))
    att = att.masked_fill(self.bias[:,:,:T,:T] == 0, float('-inf'))
    att = F.softmax(att, dim=-1)
    att = self.attn_dropout(att)
    y = att @ v # (B, nh, T, T) x (B, nh, T, hs) -> (B, nh, T, hs)
    y = y.transpose(1, 2).contiguous().view(B, T, C) # re-assemble all head outputs side by side

    # output projection
    y = self.resid_dropout(self.c_proj(y))
    return y
```

Causal Attention

```
b.attn.bias[:,:,:T,:T]
```

```
tensor([[[[1., 0., 0., 0., 0., 0.],
          [1., 1., 0., 0., 0., 0.],
          [1., 1., 1., 0., 0., 0.],
          [1., 1., 1., 1., 0., 0.],
          [1., 1., 1., 1., 1., 0.],
          [1., 1., 1., 1., 1., 1.]]]])
```

mask fill

```
def forward(self, x):
    B, T, C = x.size() # batch size, sequence length, embedding dimensionality (n_embd)

    # calculate query, key, values for all heads in batch and move head forward to be the batch dim
    q, k ,v  = self.c_attn(x).split(self.n_embd, dim=2)
    k = k.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)
    q = q.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)
    v = v.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)

    # causal self-attention; Self-attend: (B, nh, T, hs) x (B, nh, hs, T) -> (B, nh, T, T)
    att = (q @ k.transpose(-2, -1)) * (1.0 / math.sqrt(k.size(-1)))
    att = att.masked_fill(self.bias[:,:,:T,:T] == 0, float('-inf'))
    att = F.softmax(att, dim=-1)
    att = self.attn_dropout(att)
    y = att @ v # (B, nh, T, T) x (B, nh, T, hs) -> (B, nh, T, hs)
    y = y.transpose(1, 2).contiguous().view(B, T, C) # re-assemble all head outputs side by side

    # output projection
    y = self.resid_dropout(self.c_proj(y))
    return y
```

why convert 0 to -inf?

```
def forward(self, x):
    B, T, C = x.size() # batch size, sequence length, embedding dimensionality (n_embd)

    # calculate query, key, values for all heads in batch and move head forward to be the batch dim
    q, k ,v  = self.c_attn(x).split(self.n_embd, dim=2)
    k = k.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)
    q = q.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)
    v = v.view(B, T, self.n_head, C // self.n_head).transpose(1, 2) # (B, nh, T, hs)

    # causal self-attention; Self-attend: (B, nh, T, hs) x (B, nh, hs, T) -> (B, nh, T, T)
    att = (q @ k.transpose(-2, -1)) * (1.0 / math.sqrt(k.size(-1)))
    att = att.masked_fill(self.bias[:,:,:T,:T] == 0, float('-inf'))
    att = F.softmax(att, dim=-1)
    att = self.attn_dropout(att)
    y = att @ v # (B, nh, T, T) x (B, nh, T, hs) -> (B, nh, T, hs)
    y = y.transpose(1, 2).contiguous().view(B, T, C) # re-assemble all head outputs side by side

    # output projection
    y = self.resid_dropout(self.c_proj(y))
    return y
```